

The Complex Exponential Method (Prony) Application to lab measurements

The files contain the measured impulse (or free) response functions of a bridge like structure, ideally simply supported.

In each file much information is recorded but the most valuable is:

- 1) the bandwidth "Track1_TrueBandWidth" which allows to compute the sampling frequency fs
 $fs = 2.56 * \text{Track1_TrueBandWidth};$
- 2) the time histories of the various impulse responses, named "Track1", ..., "Track8".

complex exponential method

Only the decaying part of the time histories has to be elaborated by the CEM so that pre-processing of data is necessary to remove the useless parts. The procedure is here summarized:

```
>> load name01 % load the data
>> plot(Track1) % plot the time history corresponding to channel 1, for example
>> c1= Track1(n1:n2); % put in variable c1 the part of the signal which only contains a
% SINGLE free response (from point n1 to point n2)
>> c1 = c1-mean(c1); % the mean component (without any physical meaning) should be removed
>> save c1.dat c1 -ascii % save variable c1 as file c1.dat
```

Repeat for all the channels., using the SAME n1 and n2.

Files c1.dat, ..., c8.dat are now ready to be elaborated by the CEM.

Estimation of natural frequencies.

To properly select the modal frequencies it is useful to plot at least one spectrum, computed from an impulse response. In this manner computational poles may more easily be removed.

To the same purpose it is also useful to run the code prony2.m, which produces stabilization diagrams.

In this particular case it is also possible to compute the theoretical frequencies by using an Euler-Bernoulli model of the beam thus obtaining:

$$\omega_r = \frac{r^2 p^2}{L^2} \sqrt{\frac{EI}{m}}$$

Estimation of damping ratios and mode shapes.

By analyzing c1.dat, ..., c8.dat we should obtain modal damping ratios, amplitudes of oscillation and phases (mode shapes) for the first modes.

The procedure goes like this:

- 1) run prony_en.m by choosing c1.dat and a fixed number of modes "n" (model order);
- 2) take note of the values of damping, amplitude and phase corresponding to the natural frequencies above determined;
- 3) repeat steps 1) and 2) by using files c2.dat, ..., c8.dat;
- 4) compute the mean value of damping ratios and frequencies for each mode;
- 5) define the mode shapes by taking into account amplitude and phase of each measured point.